

A Self-Closing Gate Conundrum

#1 Victory Royale

Element B: Documentation and Analysis of Prior Design Attempts

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

A weight and a pulley was used often in the past as a way to close doors, when opening the door it would lift a weight that would then start pulling the door close when you stop holding it open. The good thing about this is that it is very effective at closing the door and is easy to make and install. The bad thing is that it makes the weight of the door double or more so if it is a heavy door it will be even heavier and it is not very durable. The weight and pulley would not meet our stakeholders criteria because it is too heavy and not durable enough.



A hydraulic which compresses hydraulic fluid to push the door back shut is another that was used in the past. The good thing about this is that it closes heavy doors very well while still being lightweight. The bad thing about this is that it can not be cleaned very well and it is hard to make your own hydraulic, so you have to buy it. It does not meet our stakeholders criteria because it can not be cleaned easily and we would have to buy it and not make it.



Another prior solution is having a mechanical system like an



electronic door hinge that closes the door for you. The good thing about this is that it does not stand out and can be hidden within the hinge. The bad thing about this is that it is hard to clean and fix as well as can not be built and has to be bought. This does not meet a lot of criteria for our stakeholders like it cannot be cleaned easily, and can not be made easily.

Student G is currently solving this problem by just closing the gate himself which wastes time and he sometimes forgets to close it. The only positive is that the gate will not get slammed shut and break over time. There are a lot of negatives like forgetting to close the gate and it stays open for something to get in or if student G is carrying something he has to go back to close the gate which is a huge pain.



Works Cited

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